Reading reversal and persistence: what the existing data show

Simulation 2, `task003-symmetric` and `task003-nosolution`

Step 1 of the key-phenomena study; 20 October 2026; no new simulations

Setting and notation. Main setting: $q_c = 12$, $\lambda_c = 1$, vote gate on, budget $B = 30$, window $W = 48$, 1,000 paired episodes per setting. G: paired susceptibility of the false controller (target A2), i.e. the mean over episodes of the A2 vote share with the controller minus the same episode's share without it. $D(q_1 \to q_2) = G(q_2) - G(q_1)$, computed per episode: episodes are paired across q because every random stream is keyed by seed, episode and purpose, never by q. $\delta(s)$: the paired effect s time units after the controller's last post t^*. Bars and bands: 95% intervals. Details and tables: `02_reanalysis_existing_data.md`.

1 The reading reversal

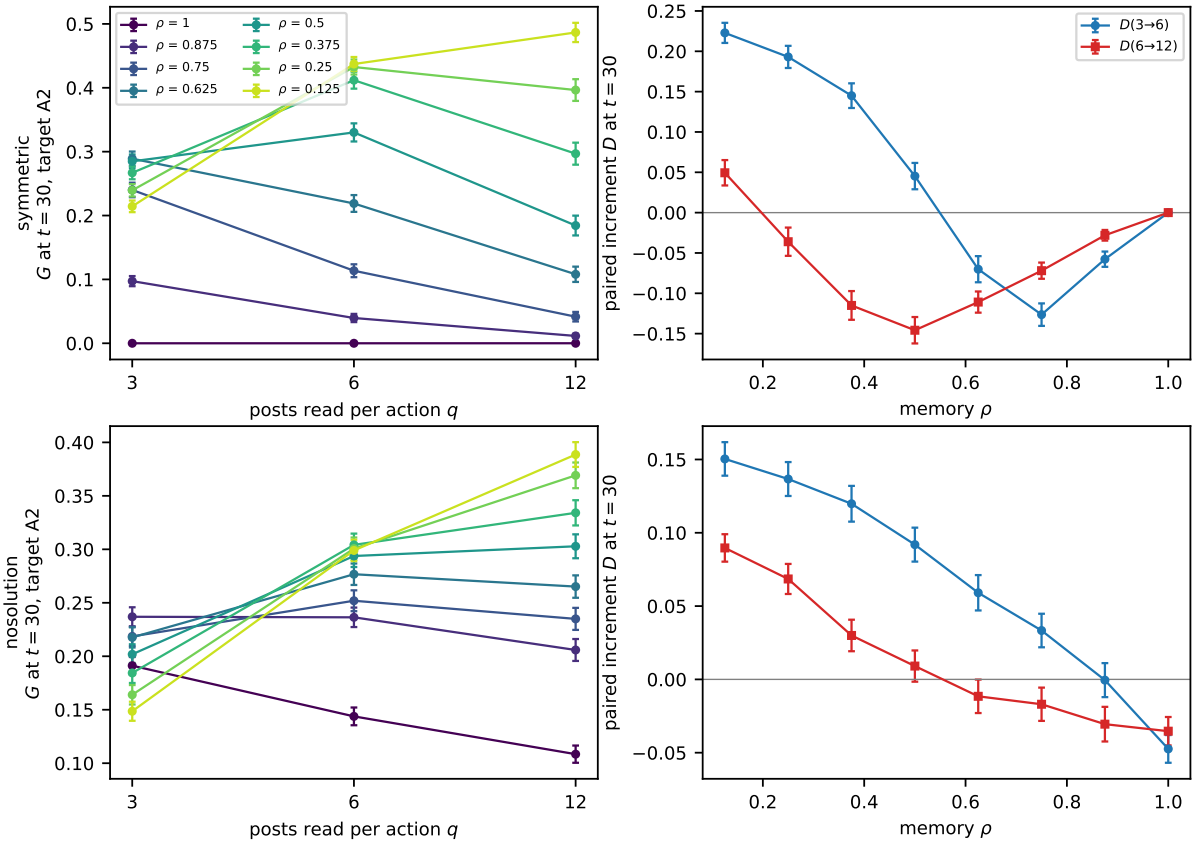


Figure 1: Left: G at $t = 30$ against the posts each agent reads, q, one line per memory ρ. Right: the paired increments $D(3 \to 6)$ and $D(6 \to 12)$ against ρ. Top: symmetric; bottom: nosolution.

- Reading more helps the false controller when agents forget and protects the swarm when they remember, in both tasks.
- Each step of q changes sign at a different memory: symmetric $\rho \approx 0.55$ $(3 \to 6)$ and ≈ 0.20 $(6 \to 12)$; nosolution ≈ 0.87 and ≈ 0.55. The crossover is a sloped contour in (ρ, q), about 0.33 higher in ρ without a proof.
- The controller's best q falls as memory grows (symmetric: 12, then 6, then 3; nosolution likewise, shifted to higher ρ).
- Nosolution at $\rho = 0.75$: G peaks at $q = 6$ ($D(3 \to 6) = +0.033 \pm 0.011$, $D(6 \to 12) = -0.017 \pm 0.011$).
- The signs hold across all 72 combinations of q_c, rate and gate, and with $W = 24$, $W = 240$ and $B = 100$, except inside the crossover cells.

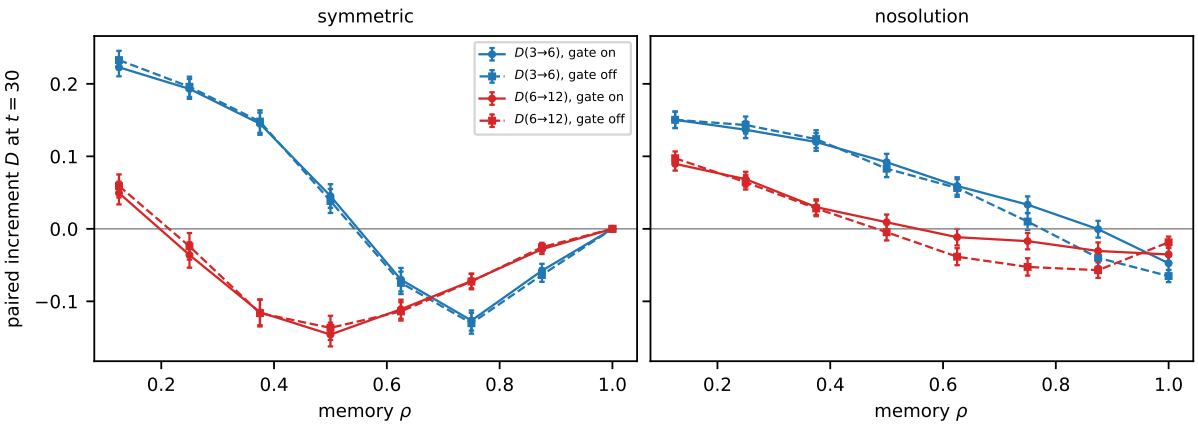


Figure 2: $D(3 \to 6)$ and $D(6 \to 12)$ at $t = 30$ with the vote gate on (solid) and off (dashed). With the gate off the controller posts 27.9 messages at every q.

- At equal dose (gate off) the increments keep their sign wherever it is clear and change by at most 0.04: the reversal is not a dose effect.

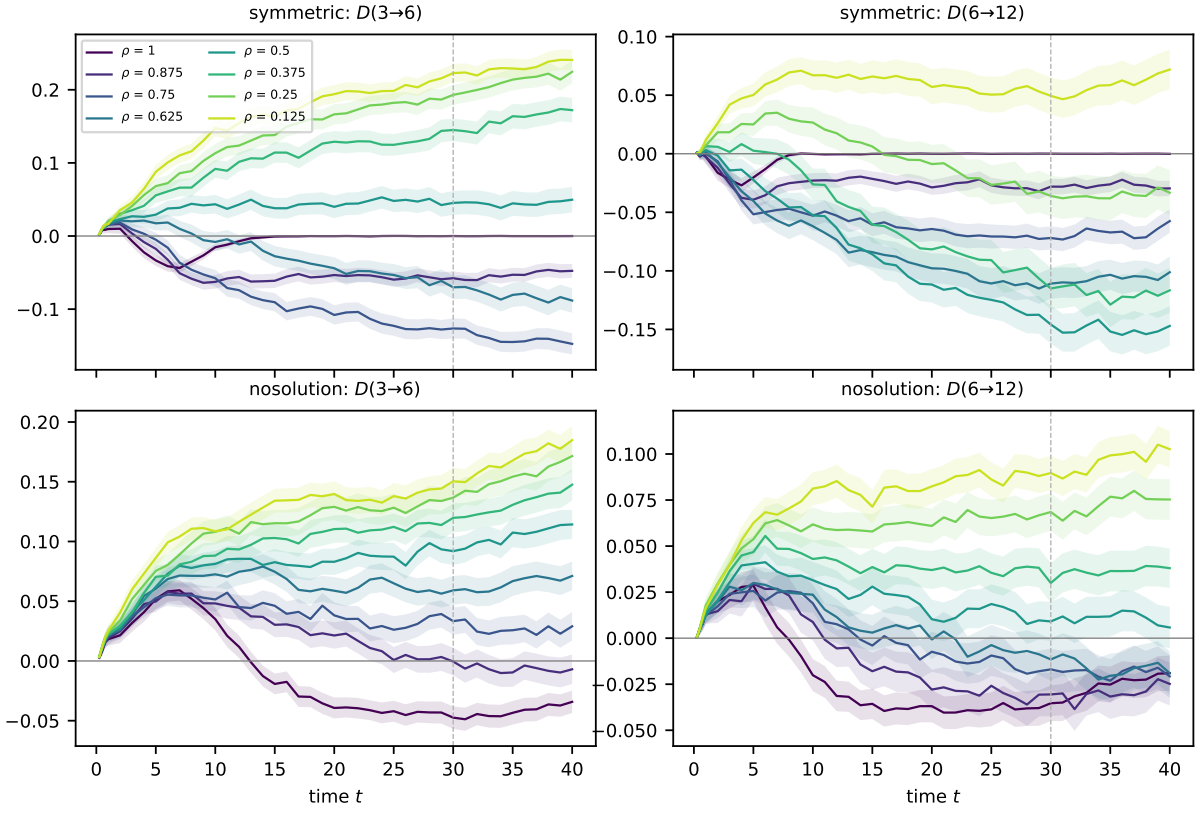


Figure 3: $D(3 \to 6)$ and $D(6 \to 12)$ over time, one line per ρ; the controller stops at $t = 30$ (dashed).

- At $t = 2$ every increment is positive (except symmetric $D(6 \to 12)$ at $\rho \geq 0.5$, ≥ -0.016): reading first spreads the controller's facts faster.
- The protective sign builds up over tens of time units, e.g. nosolution $\rho = 1$, $D(3 \to 6)$: $+0.052$ at $t = 5$, -0.047 at $t = 30$.

2 What agents remember and what the board carries

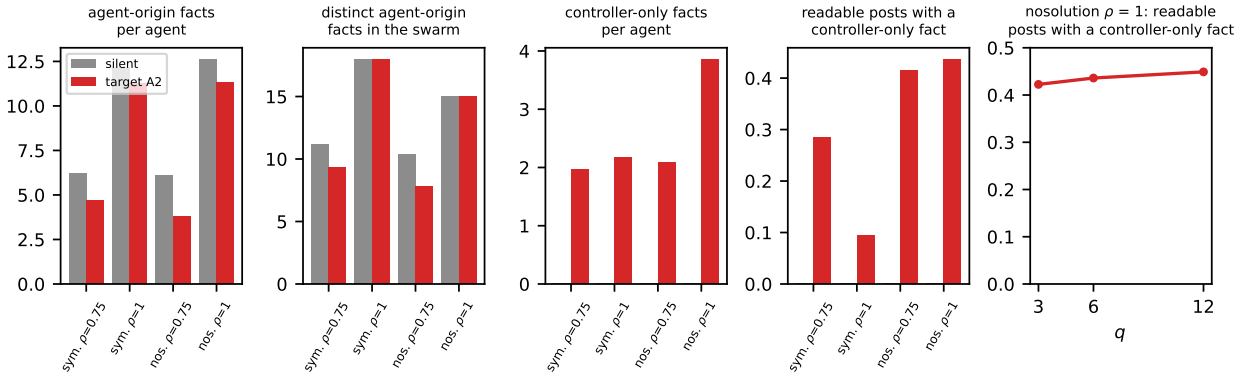


Figure 4: Full records ($q = 6$, $\lambda_c = 1$, $t = 30$): silent (grey) against the false controller (red), for symmetric and nosolution at $\rho = 0.75$ and 1. Right: nosolution, $\rho = 1$, readable posts with a controller-only fact against q (seed 2027, posts only).

- The controller crowds out the agents' own evidence: at $\rho = 0.75$ agents remember 1.5–2.3 fewer agent-origin facts and the swarm holds 1.8–2.6 fewer distinct ones.
- At $\rho = 1$ every agent fact is held by someone in both arms (18/18, 15/15); in symmetric this is a proof and $G = 0$.
- Controller-only facts make up 29–44% of readable posts, except symmetric $\rho = 1$ (10%).
- Nosolution $\rho = 1$: the controller-only share of the board rises slightly with q (0.42 → 0.45) while G falls (0.19 → 0.11). The protection is not on the board; memory records across q are needed (step 2).

3 Persistence after the controller stops

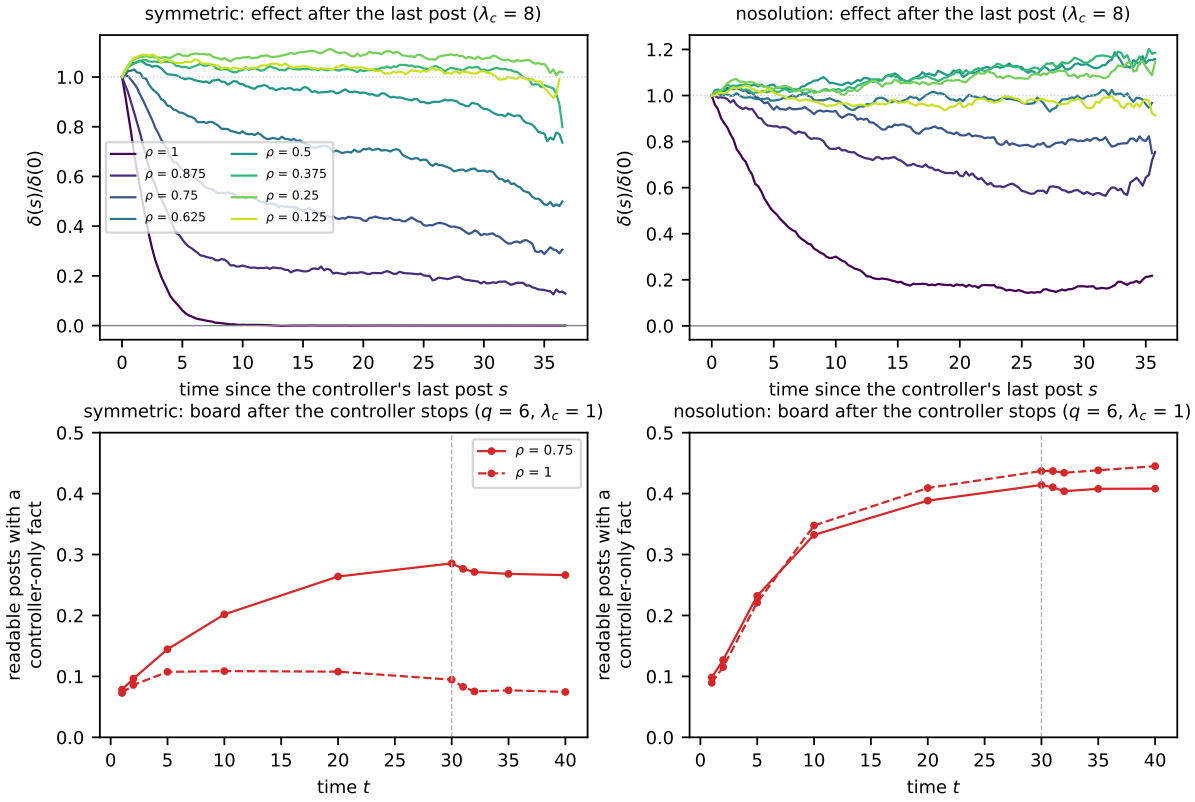


Figure 5: Top: $\delta(s)/\delta(0)$ after each episode's last post, $\lambda_c = 8$ (median t^* 4–6 symmetric, 9–10 nosolution), one line per ρ; censored at $t = 40$. Bottom: readable posts with a controller-only fact, false arm, $q = 6$, $\lambda_c = 1$.

- For $\rho \leq 0.5$ the effect does not decay for more than 35 time units after the last post, in both tasks ($\delta(30)/\delta(0) = 0.87$–$1.14$); an unread fact's half-life is 0.3–1 time unit.
- At higher memory it relaxes partly to a plateau: symmetric $\rho = 0.75$ to about 0.37 (half-time 11.5), nosolution $\rho = 1$ to about 0.17 (half-time 5). Whether the plateaus are permanent is unknown (no data beyond $t = 40$).
- With a proof and full memory the effect vanishes within 2 time units.
- From $t = 30$ to 40 the controller-only share of readable posts hardly changes (nosolution $\rho = 0.75$: 0.41 $\rightarrow$ 0.41): agents keep reposting the controller's facts.

4 Backfire (secondary)

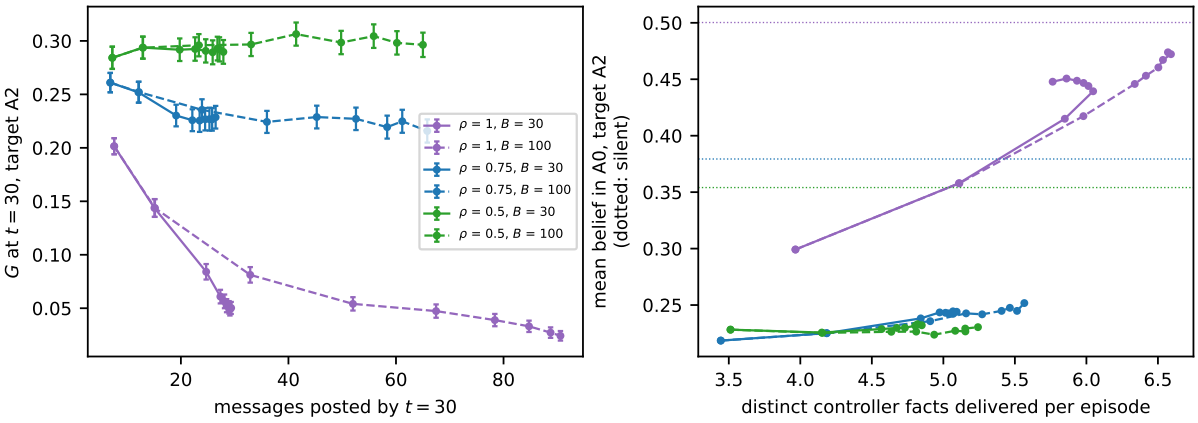


Figure 6: Nosolution, $q = 6$, gate on, $t = 30$, budgets 30 (solid) and 100 (dashed), rates 0.5–8. Left: G against messages posted. Right: the false arm's mean belief in A0 against distinct controller facts delivered (dotted: silent arm).

- At $\rho = 1$, more messages lower G (0.20 $\rightarrow$ 0.02 at 91 messages) as agents' belief in A0 returns toward the silent 0.50 (0.30 $\rightarrow$ 0.47).
- At $\rho = 0.75$ the backfire is weak (0.26 $\rightarrow$ 0.22); at $\rho = 0.5$ G is flat in messages.

Open. Why reading protects (memory composition across q at $\rho \leq 0.5$: step 2) and what carries the persistence (facts, biased state or communication: step 3).